

Lecture 23

3/13/26

JRA

Discussing two SR techniques that won NP

PALM - stochastic

Cell w/ Filamentous structures w/ Fluorophores attached

Assume Fluorophores dark at first

PA GFP

H.t w/ weak activation light 405nm

Activate w/ light
Weak enough so only a few fluorophores are activated - non overlapping fluorophores

Collect data from the activated Fluorophores,
computer tabulates centers, Fluorophores bleach,
REPEAT again & again green pink blue

Repeat up to 10 000 x

Stochastic switching, coordinate readout,
ultimately build up an image

Image via "pointillism"

There are some subtleties

Center is calculated

Need enough signal to determine center well

$$\text{uncertainty} = \frac{\text{PSF width}}{\sqrt{m}}$$

$\sqrt{m}$
of collected photons

ex) If PSF width ≈ 200 & $m = 400$

$$\text{Uncertainty} = \frac{200}{\sqrt{400}} = 10 \text{ nm}$$

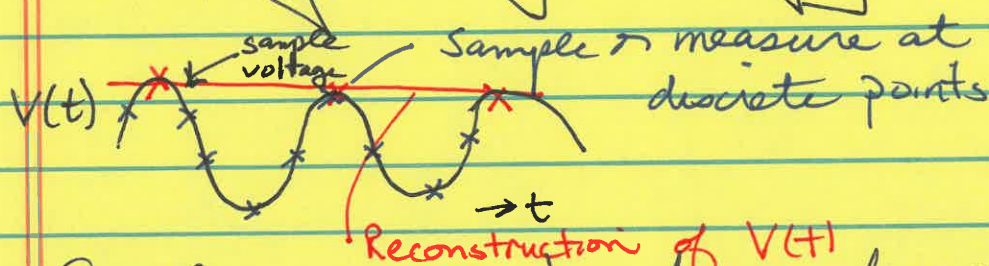
↑ # of collected photons

Dirty little secret of PALM

Pervasive
issue
in imaging

What's this?

Sampling in Digital Imaging

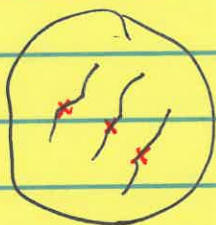


Sampling must be done frequently enough to reproduce the signal accurately

Bad: Reconstruct sinusoidal signal as flat

This can cause big issues w/ PALM

PALM



Sample too sparsely
 $\Rightarrow$ DOES NOT show filaments

To sample enough, need a LOT of repetitions
 10000x

Sampling Theorem tells how to sample correctly

Ex 1 Let's say you want to achieve diffraction limited resolution w/ normal Fluorescence e.g. LSCM

$$R = 200 \text{ nm}$$

Want 200 nm resolution $\rightarrow$ won't get unless sample enough $\Rightarrow$ sample size required $d = \frac{R}{2}$

For std Fluorescence $d = 100 \text{ nm}$

This is like pixel size. Cameras are adjusted
 $d = R/2$ sample size required $d = 100 \text{ nm}$

STED -

Strengths - see blackboard photos

Processing (to find centers)

Can use cameras

→ targeted switching & coordinate readouts

Similar to LSCM - scanning technique



Spot is targeted to certain spot in cell
 Know location Computer tabulates center
 position of spot r_i
 i th spot

Know position & location.

What's the advantage over LSCM?

STED ^{scanning} spot is ~ 10-fold smaller than an
 LSCM spot

This is like painting w/ a finer brush

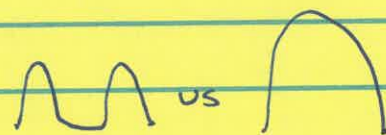


VS —

Laser spot Big - Fluorescence at same time $\Rightarrow$ unresolvable

STED spot

Bright - Dark - Bright



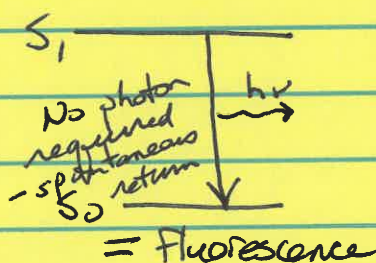
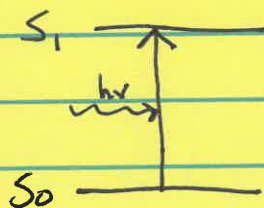
How do you make small spot?

Stimulated Emission is the key to reducing spot size.

Stim absorption
Requires photon

spont.
emission

New kid
Stim emission



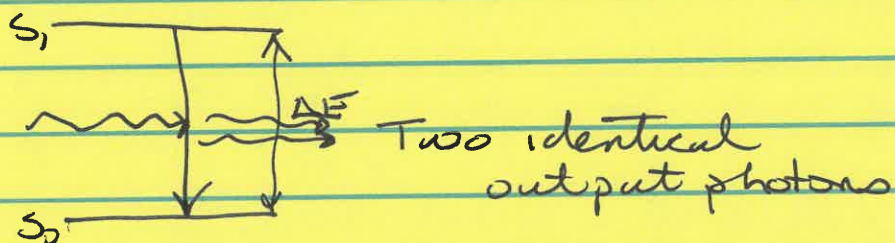
Needs photon to stimulate absorption

MRI - everything
aside stimulated

There is another form of radiation called stimulated emission

Predicted by Einstein

LASER basis



Send in photon w/ energy matching energy between states — knock electron down out of excited state

stim emission & STED

Focus now on STED application

Why 2 output photons?

1 is energy of input photon

1 is energy loss going to ground state

2 output photons original & one from energy loss identical in every way

Engineering a small spot w/ stimulated emission. (1) Need two lasers.

(a) First laser excites fluorescence

Need some fluorescence to give signal



Big spot - size of LSCM spot
to excite fluorescence

Donut shaped spot to stimulate emission
to knock out of excited state
w/ stim emission in periphery
only

Fluorophores in out depleted area no longer
fluorescent - can't fluoresce

Leave a small unquenched spot to scan with

(2) Stimulated emission knocks fluorophore
out of excited state so they don't
fluoresce

Jargon: Quench Fluorescence in periphery
of larger scanning spot
Need donut-shaped beam
using a D shaped beam

Do need two lasers

Green to excite

Red to quench

Stim energy photon is matched to emission transition
→ red shifted

23-8

Stokes shift says emission is lower E
 $\Rightarrow$ red shifted

Use longer wavelength quenching beam

One more subtlety

This must occur superfast.

stim emission when fluorophore in excited state

Molecule excited for ~ 1 ns
Hit while fluorophore still in excited state

Fluorophore must still be excited to be quenched by SE

Fluorescence lifetime is ~ 1 ns
Stimulated w/ donut in $\ll 1$ ns eg 100 psec

Requires 2 lasers, really fast

STED advantages

① Much faster than PALM

can do live specimen imaging

MOVE ② No image processing

③ Theoretically unlimited resolution
~20 nm

→ ④

Weaknesses

① 2 lasers for every fluorophore Hard for multiple fluorophores

② Very high light intensities Photo B
Photo T

③ Tricky set ups

Microscopy concepts Monday